

Abhishek's Vehicle Portfolio - Project Summary

Project Type: Fleet Management System

Architecture: Frontend (HTML/CSS/JS) + Backend (Node.js/Express) + MySQL Database

Folders and key files:

Frontend:

- index.html: login page and role selection.
- dashboard.html: main dashboard showing fleet overview and cards.
- add-vehicle.html: form to add new vehicle details.
- vehicle-detail.html: single vehicle details with tabs for trips, diesel, service, and docs.
- trips.html, diesel.html, service.html, finance.html, analytics.html, uploads.html: operational pages.
- driver-dashboard.html: driver-specific dashboard and job management.
- driver-register.html: driver registration form with document uploads.
- hire-drivers.html: owner page to browse and hire drivers.
- css/: contains style.css, login.css, dashboard.css for layout and theme.
- js/: contains auth.js, config.js, dashboard.js, vehicle.js, trips.js, diesel.js, service.js, realtime.js, finance.js.

Backend:

- server.js: Express server entry point, CORS config, route mounting, health check.
- config/db.js: MySQL connection pool configuration and connection test.
- middleware/auth.js: JWT token verification and optional auth handler.
- routes/: auth.js, vehicles.js, trips.js, diesel.js, service.js, finance.js, analytics.js, uploads.js, drivers.js, jobs.js, reviews.js, compliance.js, ocr.js.
- database/setup.js: script to create database schema and initial tables.
- backend/.env: environment variables for DB credentials, JWT secret, frontend URL (not committed).

Workflow:

1. User opens the frontend login page and submits credentials.
2. Frontend sends credentials to backend `/api/auth/login`.
3. Backend verifies user and returns JWT.
4. Frontend saves JWT locally and includes it in Authorization headers for API calls.
5. Backend routes validate JWT using `middleware/auth.js` and perform database queries using `config/db.js`.
6. Users manage vehicles, trips, diesel, service, compliance, and uploads via dedicated frontend pages.
7. Uploaded files are stored in `backend/uploads` and served via Express static middleware.
8. Analytics and finance pages aggregate data and render charts for reporting.

Deployment notes:

- Frontend can be hosted as static files on AWS S3.
- Backend can be deployed on AWS EC2 with Node.js and MySQL.
- Update `frontend/js/config.js` `API_BASE_URL` to the deployed backend URL.

Generated from existing project files in vehicle-portfolio.